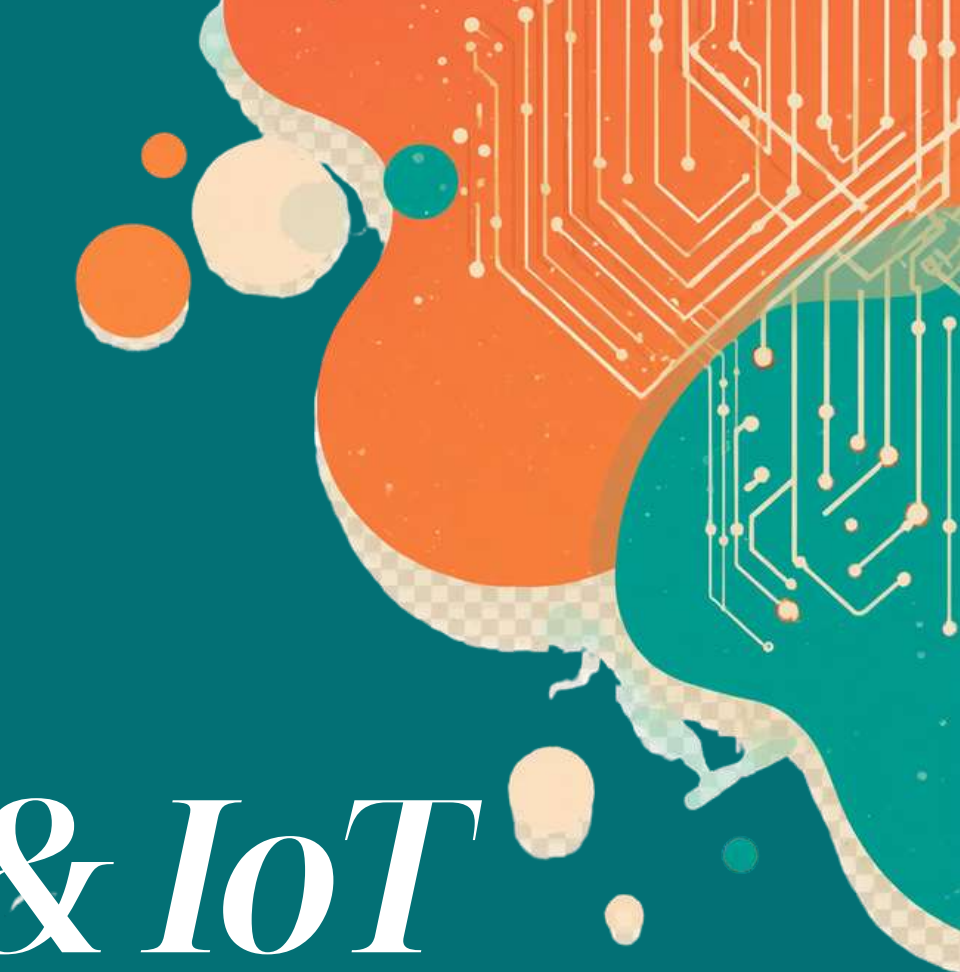
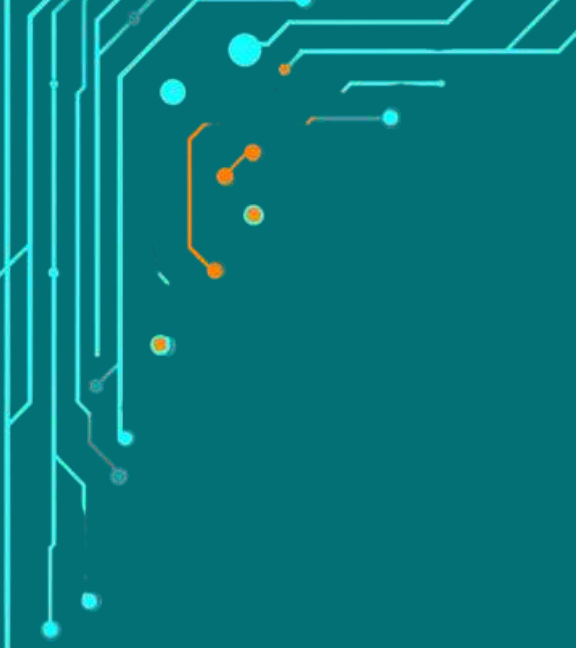




Module 1 – Introduction to Arduino & IoT

Introduction to Smart Electronics with Arduino





What Is a Smart System?

Regular System

Embedded System

IoT System

Embedded System vs IoT Device

Microcontroller + Sensors

- Standalone device
- Local control only
- No internet needed
- Reads inputs & controls outputs
- Great for simple automation
- Self-contained projects

Microcontroller + Sensors + Internet

- Connected to the cloud
- Remote control from anywhere
- Shares data online
- Part of the IoT network
- Smart & scalable systems
- Monitor via phone or web

Automation Around Us

🚦 Traffic Lights – Automatically cycle through colors to manage traffic flow and reduce accidents without human control.

🛗 Elevators – Sense floor requests, decide the most efficient route, and move passengers automatically.

🚪 Automatic Doors – Detect approaching people using sensors and open or close without any manual effort.

⚙️ Washing Machines – Follow programmed cycles to wash, rinse, and spin clothes automatically based on settings.

How Smart Systems work

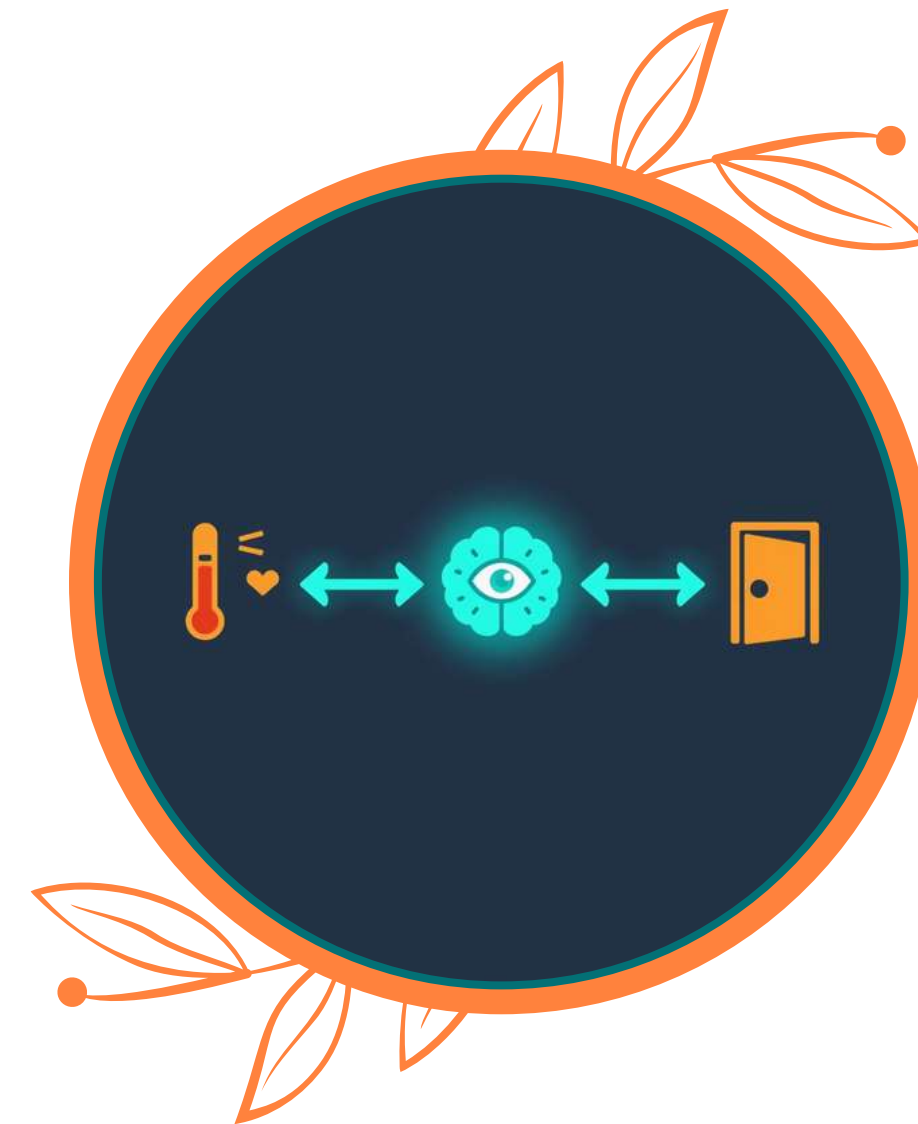
📡 INPUT → ⚙️ PROCESSING → 📡 OUTPUT

Sensors detect the world → Arduino decides what to do → Devices take action!

● Sensors + Decision Making + Action

🌡️ Thermostat: senses temperature → decides → turns heater ON

🚪 Automatic Doors: sense presence → decide → open automatically

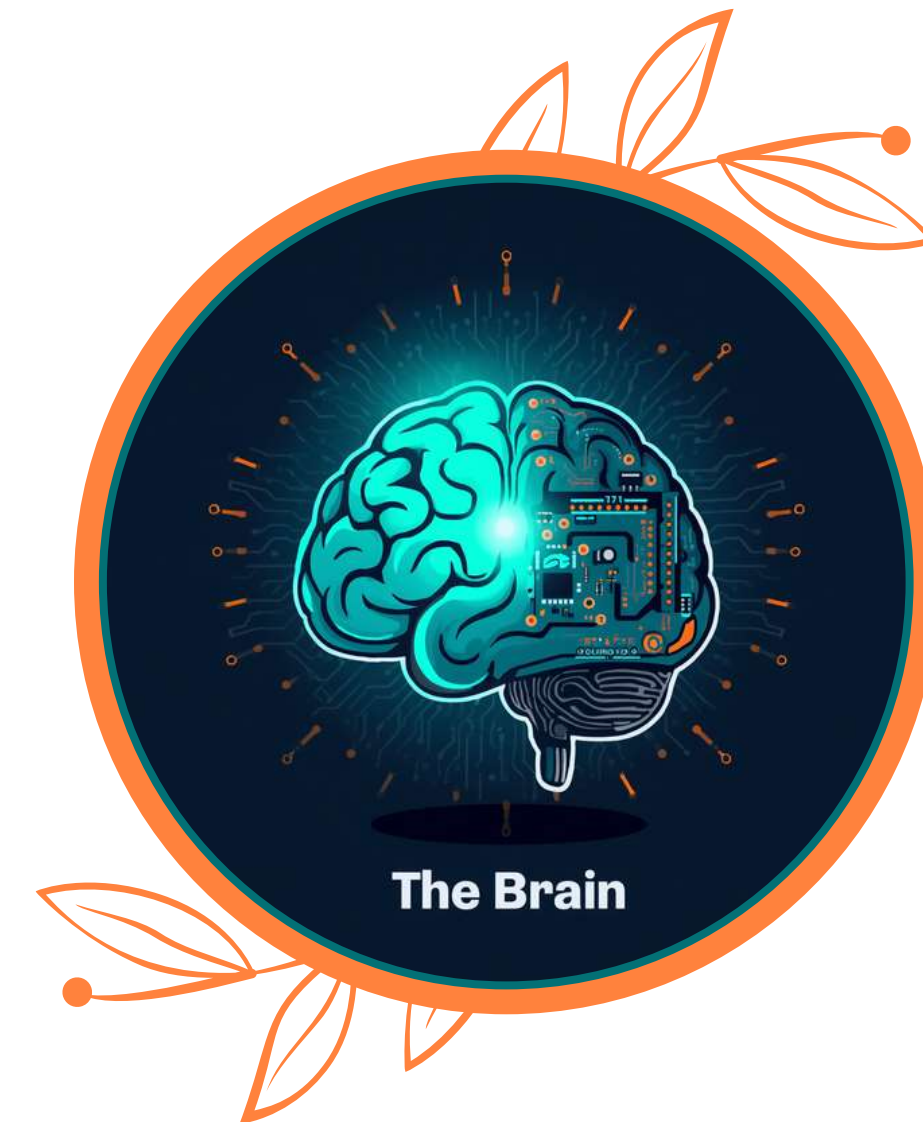


Input Devices

- 🕒 Button – Detects a press (ON/OFF signal); like a light switch telling Arduino "go!"
- 🔲 Potentiometer – A rotating knob that sends a variable signal; controls speed, brightness, or volume
- 💡 LDR (Light Sensor) – Detects light levels; gets brighter → resistance drops → Arduino reads the change!
- 🗻 Ultrasonic Sensor – Measures distance using sound waves; like a bat's echolocation!

Processing Unit

- 🧠 Microcontroller = The Brain
 - Reads signals from input devices
 - Makes decisions based on code
 - Controls output devices



Output Devices

- 💡 LEDs (Lights) – Visual signals and indicators; light up to show status or alerts
- 🔔 Buzzers (Sounds) – Audio alerts and tones; beep or buzz to signal events
- ⚙️ Servo Motors (Movement) – Precise rotation and motion; move parts, open gates, rotate objects



Example 1 – Button → LED ON

Press the Button → Arduino Detects Input

A physical button is connected to the Arduino. When you press it, the Arduino reads the signal as a HIGH input on its digital pin. This is how the system senses a real-world action!

Arduino Processes → LED Turns ON

The Arduino runs a simple program: if the button is pressed, send power to the LED pin. The LED lights up instantly! This is the complete Input → Processing → Output cycle in action.



Example 2 – Darkness → Light ON

"Darkness Detected" → LDR sensor reads low light levels → Arduino receives the signal and processes the input data

💡 Fun Fact: LDRs are used in street lamps that automatically turn ON at night!

"Arduino Processes" → Checks if light level is below threshold → "Light ON" → Arduino sends signal to LED/bulb → Light activates automatically! ☀️



Example 3 – Object Near → Alarm

Object Detected Nearby → Arduino Triggers Alarm

The ultrasonic sensor measures the distance to an approaching object. When the object gets too close, it sends a signal to the Arduino board, which processes the input and decides to activate the alarm output.

Buzzer Sounds! 🔔

The Arduino triggers the buzzer, which emits a loud alert sound. This is the same principle used in parking sensors, security alarms, and obstacle-avoidance robots!



Meet Arduino

What it is: An open-source microcontroller board that reads inputs and controls outputs.

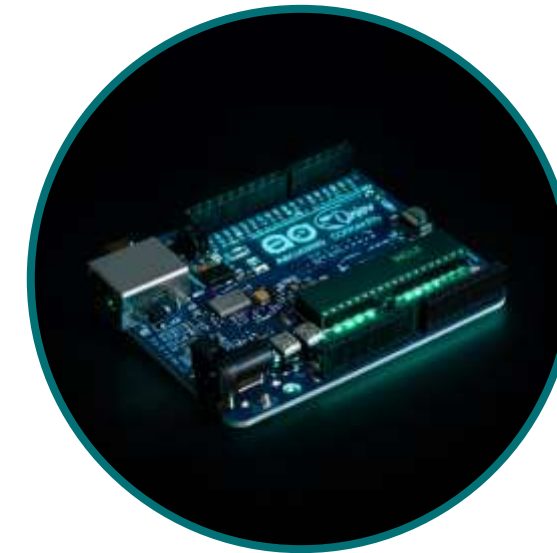
Why makers love it:

- Easy to learn & program
- Versatile for any project
- Fun and beginner-friendly

Types:

- Arduino Uno
- Arduino Nano
- Arduino Mega

💡 Fun Fact: Arduino is named after a bar in Italy where the founders used to meet!

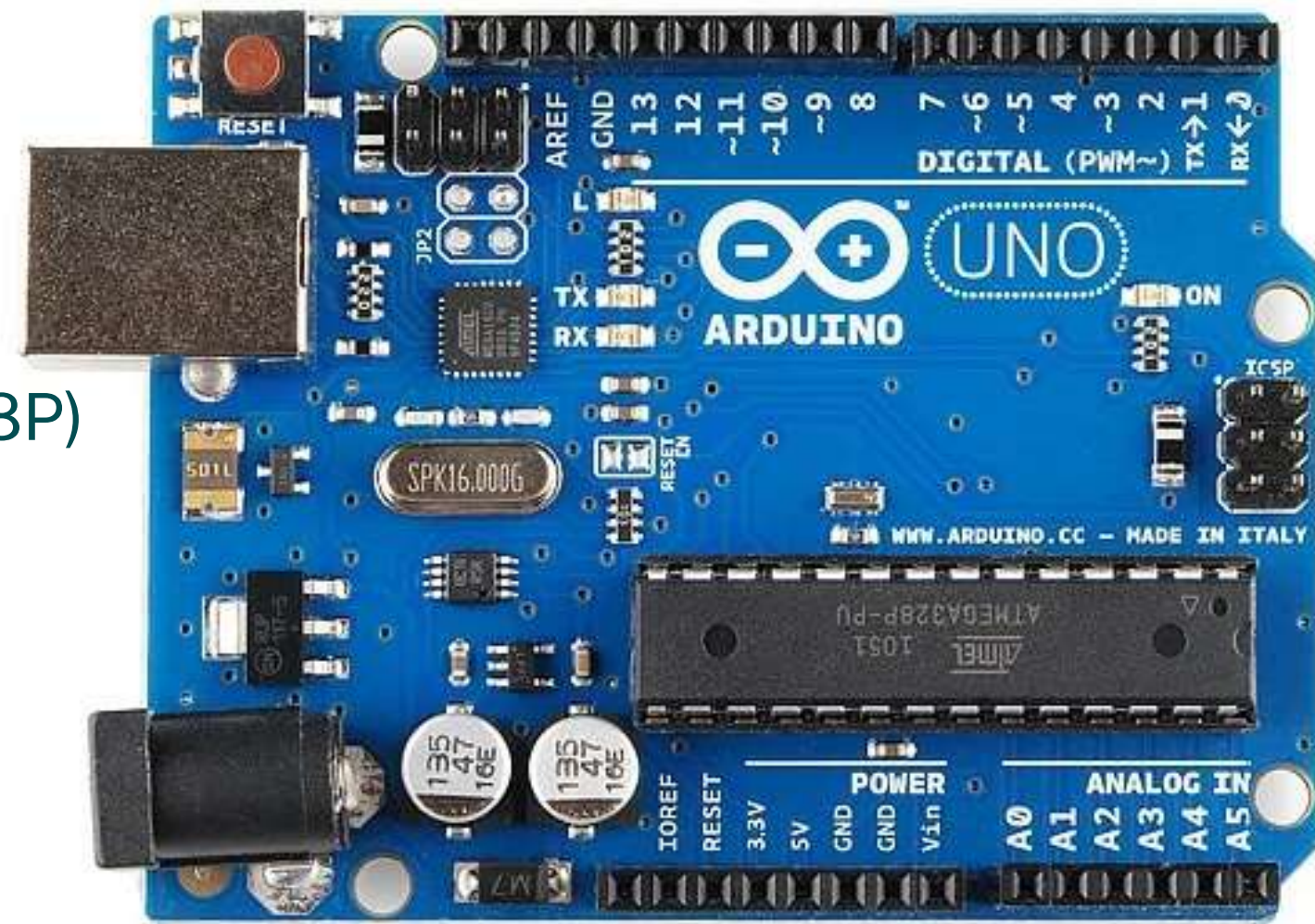


Applications of Arduino

-  Robotics – Build autonomous robots and smart machines
-  Home Automation – Control lights, locks, and appliances remotely
-  Agriculture – Automate irrigation and monitor soil conditions
-  Scientific Research – Collect and analyze real-world sensor data
-  Space Technology – Power experiments in satellites and rovers

Arduino Uno Board Tour

- 🔌 USB Port – connects to your computer
- ⚡ Power Jack – external power supply
- 🧠 Microcontroller – the brain (ATmega328P)
- 📌 Digital Pins – ON/OFF signals (0–13)
- 📊 Analog Pins – variable signals (A0–A5)
- 🔄 Reset Button – restart the program



Components We Will Use

Input

-  Button – detects a press
-  Potentiometer – adjustable knob
-  LDR – senses light levels
-  Ultrasonic Sensor – measures distance

Output

-  LED – lights up as output
-  Buzzer – makes a sound
-  Servo Motor – moves and rotates
-  Breadboard – connects components (no soldering!)
-  Jumper Wires – link everything together

Workshop Safety Rules

✓ DO's:

- Handle components gently
- Keep liquids away from electronics
- Disconnect power when wiring
- Double-check connections before powering on
- Ask for help when unsure

✗ DON'Ts:

- Touch powered circuits with bare hands
- Use damaged or frayed wires
- Rush connections — take your time!
- Leave components loose on the board
- Work alone on powered setups



What You Will Build

Reflex Timer Game – Test your reaction speed with buttons and LEDs!

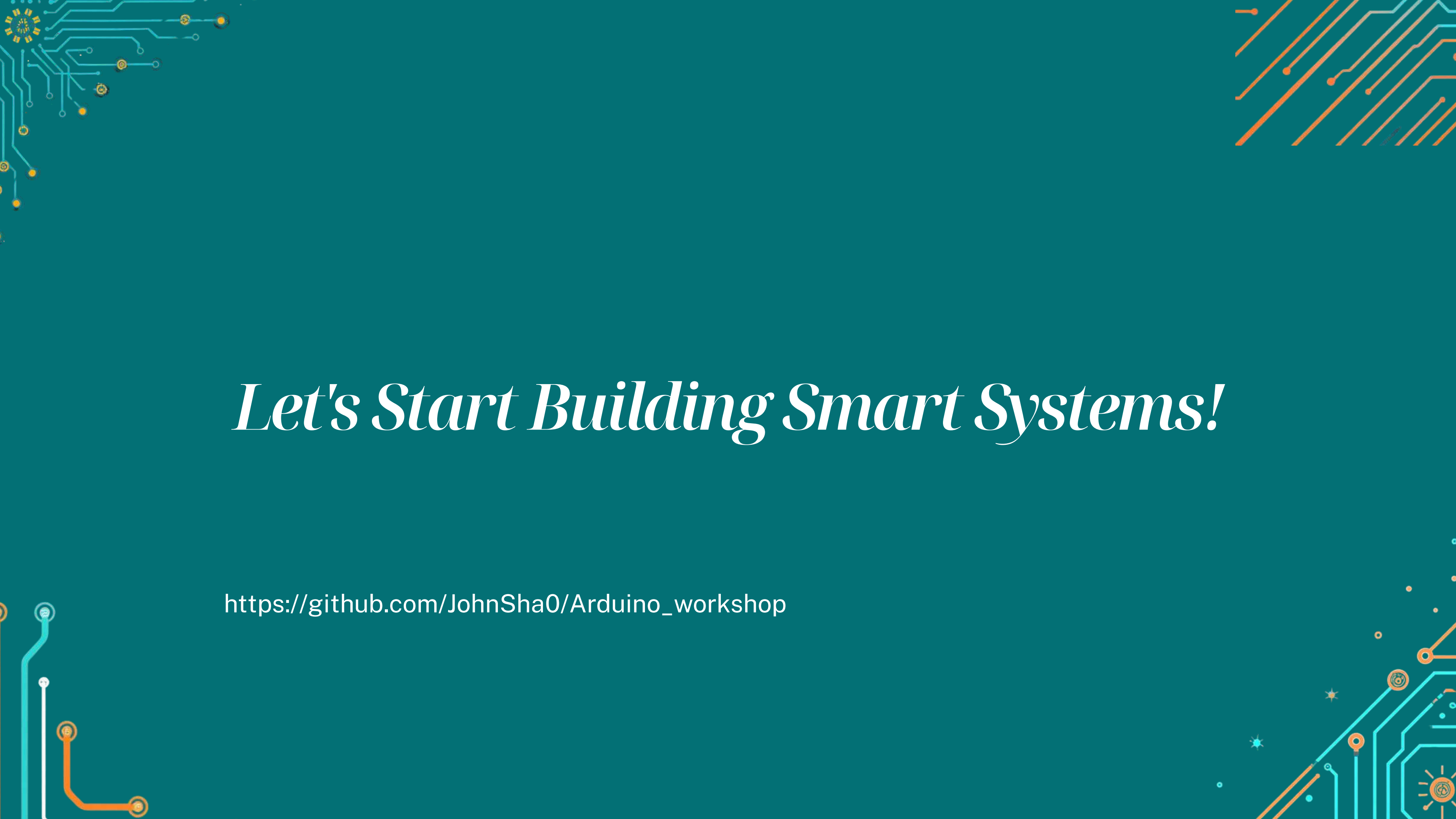
Smart Lighting System – Automatically turn lights on/off based on brightness.

Parking Alert System – Detect objects with ultrasonic sensor and trigger a buzzer.

Intrusion Alarm – Sense movement and sound alarm.

Automatic Barrier Gate – Sense movement and control a servo motor gate.



The background is a solid teal color. In the corners, there are decorative circuit-like patterns. The top-left corner features a complex network of light blue and yellow lines with small circular nodes. The top-right corner has a series of parallel orange and yellow lines. The bottom-left corner shows a few vertical and horizontal lines in light blue and orange. The bottom-right corner is more intricate, with orange and yellow lines, small star-like shapes, and a circular node resembling a microchip.

Let's Start Building Smart Systems!

https://github.com/JohnSha0/Arduino_workshop